

Control Structures

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Introduction

In this lab you will implement a simple calculator on the target hardware. The operands can be entered through the dip switches and the operation can be selected through the rotary switch.

Learning Objectives

- you can implement a switch-case statement with a jump table in assembly
- you strengthen your skills in the application of arithmetic and logic instructions

Task 1

Implement a program, that reads 8-bit values from the DIP-switches and performs different operations on them. The result shall be displayed on the LEDs. Use the given project frame.

- Read the comments in the given program frame. They will show you where to enter your code.
- Implement the switch-case statement based on the lecture slides.

The program shall meet the following requirements:

- The first operand (op1) shall be entered through the DIP-switches `S[15..8]`.
- The second operand (op2) shall be entered through the DIP-switches `S[7..0]`.
- The 8 bit operands shall be zero-extended to 32 bit.
- The operation shall be selected through the rotary switch, according to the table below.
- The position of the rotary switch shall be displayed on the 7-segment display. The operands 1 and 2 shall be displayed on `LED[15..0]` (above the corresponding DIP-switches).
- The 16 bit result shall be displayed on `LED[31..16]`.
- A logical '1' shall mean that the corresponding LED is on.

- The program shall use a jump table.

Hex-Switch	Operation
0x00	all 16 LEDs off
0x01	op1 + op2
0x02	op1 - op2
0x03	op1 * op2 (unsigned)
0x04	op1 & op2 (AND)
0x05	op1 op2 (OR)
0x06	op1 ^ op2 (XOR)
0x07	!op1 (NOT)
0x08	!(op1 & op2) (NAND)
0x09	!(op1 op2) (NOR)
0x0A	!(op1 ^ op2) (XNOR)
0x0B	all 16 LEDs on
0x0C	all 16 LEDs on
0x0D	all 16 LEDs on
0x0E	all 16 LEDs on
0x0F	all 16 LEDs on

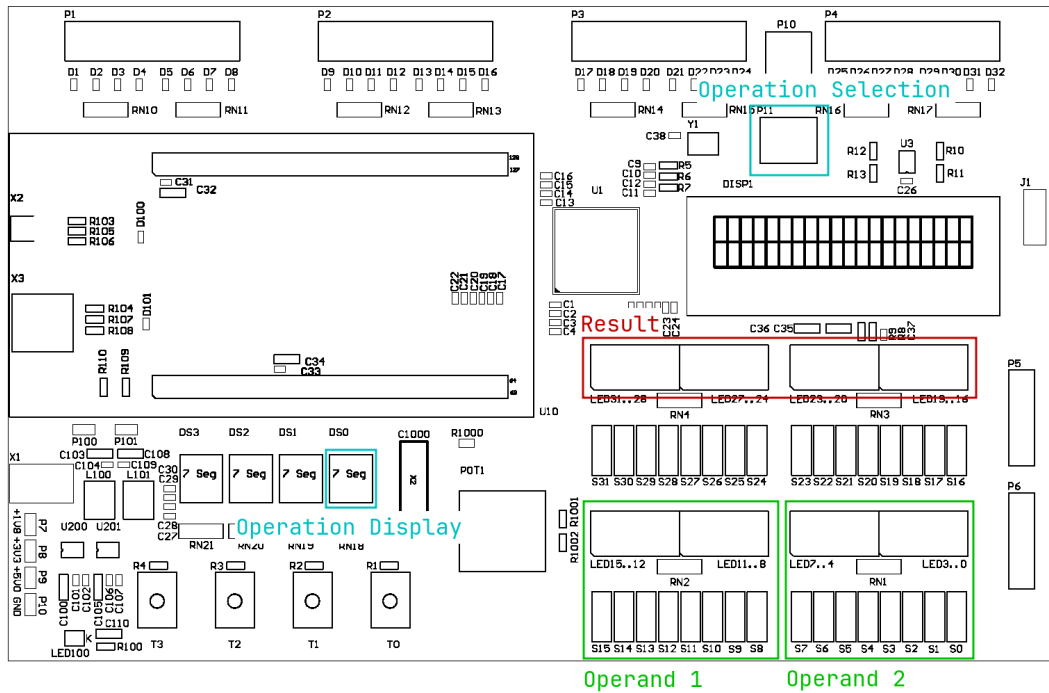


Figure 1: in- and output of operands, operation and result

Grading

The working program has to be presented to the lecturer. The student has to understand the solution/source code and has to be able to explain it to the lecturer.

Criteria

Weight

The program meets the required functionality.

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